Cascading Style Sheets (CSS)

What is CSS?

- CSS ("Cascading Style Sheets") determines how the elements in our HTML documents are displayed and formatted.
- By using CSS, we separate the content of a web page from the presentation (format and styling) of that content.
- CSS enables us to make all pages of our website look similar and consistent.
- The power of CSS is that it allows us to make site-wide formatting changes by making edits to a single file.

Three Ways to Use CSS

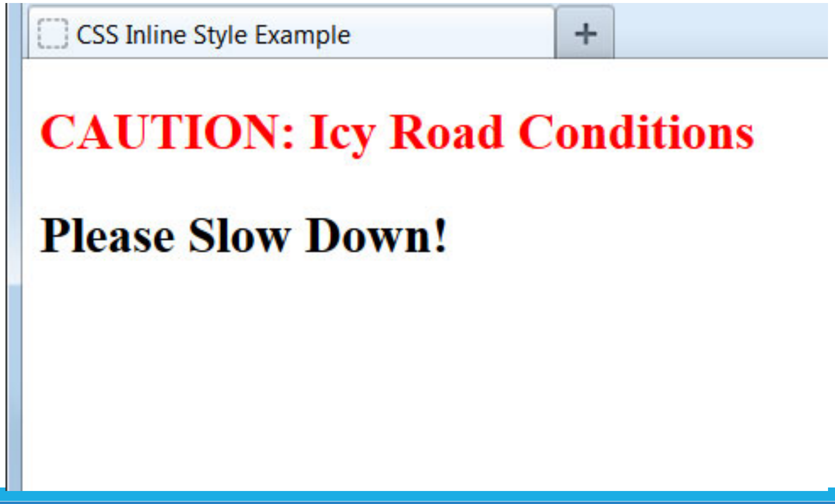
We can add CSS code in any combination of three different ways:

1. Inline Style - CSS code is placed directly into an HTML element within the <body> section of a web page.
2. Internal Style Sheet - CSS code is placed into a separate, dedicated area within the <head> section of a web page.
3. External Style Sheet - CSS code is placed into a separate computer file and then linked to a web page.

Inline Style

To define an inline CSS style, we simply add the **style** attribute to an HTML element with the CSS declaration as the attribute value:

```
<h2 style="color:red;">CAUTION: Icy Road Conditions</h2>
<h2>Please Slow Down!</h2>
```

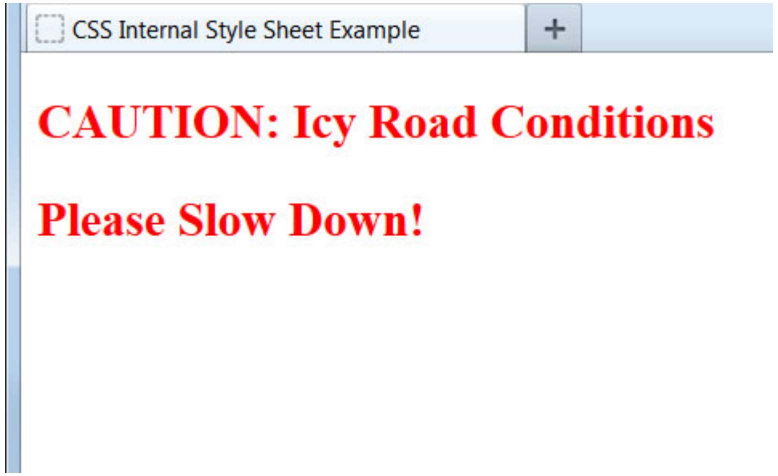


An inline style declaration is highly specific and formats just one element on the page. No other elements, including other <h2> elements on the page, will be affected by this CSS style.

Internal Style Sheet

To use an internal CSS style sheet, we add a <style> section within the <head> of the page. All our CSS declarations go within this section:

```
<head>
  ...
  <style type="text/css">
    h2 {color:red;}
  </style>
</head>
<body>
  <h2>CAUTION: Icy Road Conditions</h2>
  <h2>Please Slow Down!</h2>
</body>
```



Styles declared in the internal style sheet affect all matching elements on the page. In this example, all <h2> page elements are displayed in the color red.

External Style Sheet

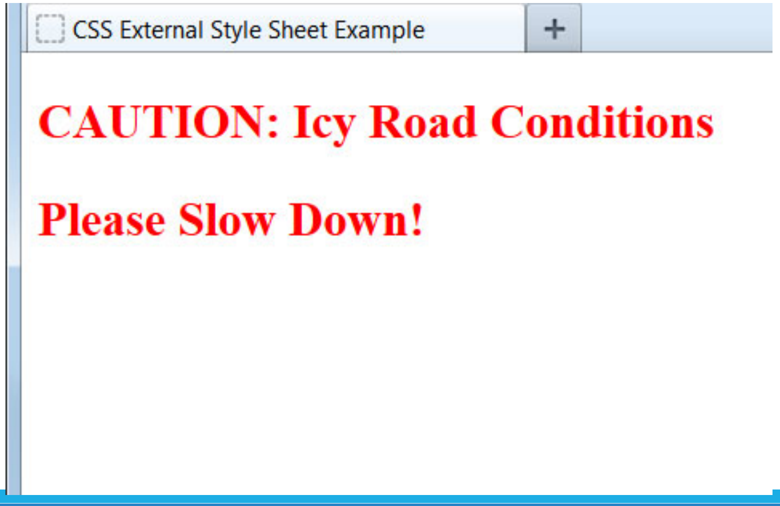
To use an external CSS style sheet, we create a new file (with a .css extension) and write our style declarations into this file. We then add a <link> element into our HTML file, right after the opening <head> tag:

style.css (separate file):

```
h2 {color:red;}
```

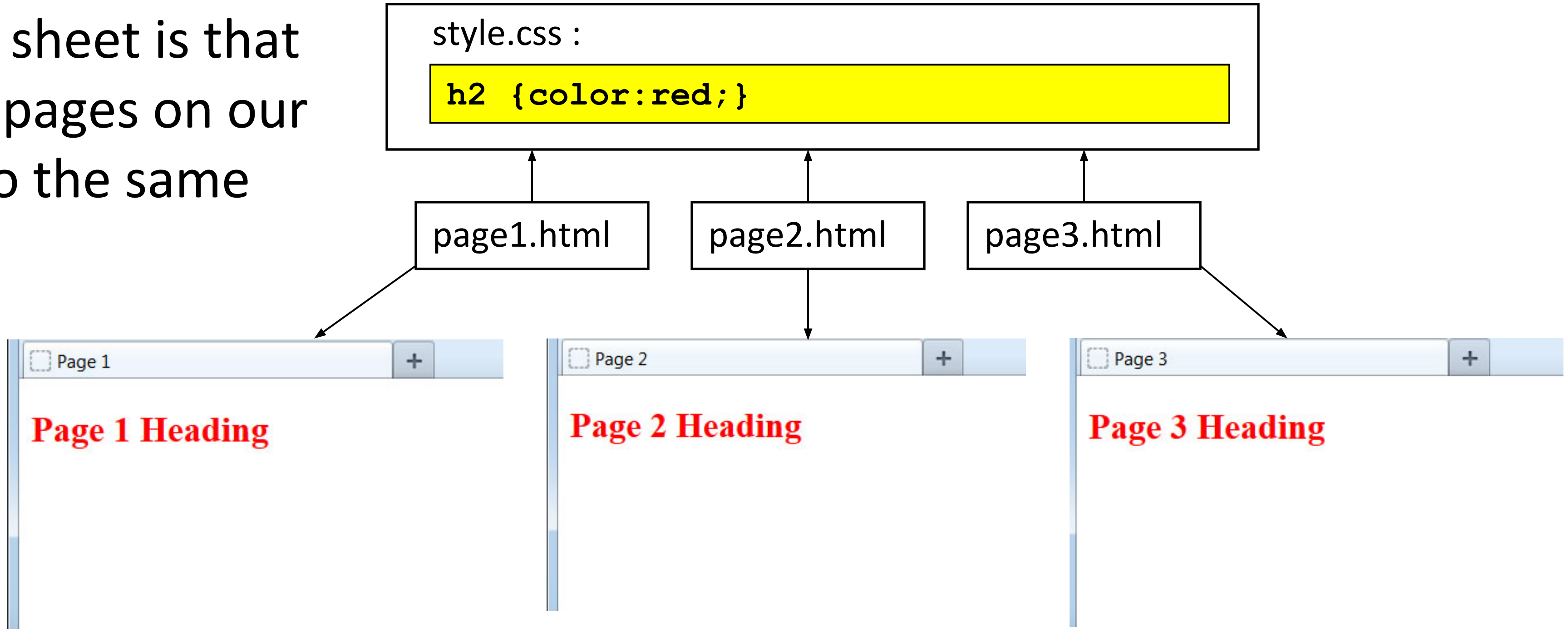
example.html file:

```
<head>
  <link rel="stylesheet" type="text/css"
   href="style.css" />
  ...
</head>
<body>
  <h2>CAUTION: Icy Road Conditions</h2>
  <h2>Please Slow Down!</h2>
</body>
```



Benefit of External Style Sheet

The real power of using an external style sheet is that multiple web pages on our site can link to the same style sheet:



Styles declared in an external style sheet will affect all matching elements on <u>all web pages that link to the style sheet</u>. By editing the external style sheet, we can make site-wide changes (even to hundreds of pages) instantly.

Internal vs. External Style Sheets

Internal Style Sheets:

- are appropriate for very small sites, especially those that have just a single page.
- might also make sense when each page of a site needs to have a completely different look.

External Style Sheets:

- are better for multi-page websites that need to have a uniform look and feel to all pages.
- make for faster-loading sites (less redundant code).
- allow designers to make site-wide changes quickly and easily.

CSS Terminology and Syntax:

Now let's take a closer look at how we write CSS code. The correct syntax of a CSS declaration is: **selector {property:value;}**

```
p {color:red;}
```

selector

property

value

Internal and external style sheets use this identical CSS syntax. Internal style sheets must use the opening and closing <style> tags to surround the CSS code, while external style sheets do not use the <style> element.

A semicolon must be placed after each CSS declaration. Omitting this semicolon is the single most common mistake made by those learning CSS.

Setting Multiple Properties

We can define as many properties as we wish for a selector:

```
p {color:red;font-style:italic;text-align:center;}
```

In this example, all text within paragraph elements will show in red italics that is centered on the page.

```
p {
  color: red;
  font-style: italic;
  text-align: center;
}
```

Just as with HTML, browsers ignore space characters in CSS code. Many designers take advantage of this fact by placing the opening and closing curly brackets on their own dedicated lines. Each of the property and value pairings are placed on their own indented line, with a space after the colon. This makes the code far easier to read.

CSS Text Properties

The following properties can be specified for any element that contains text, such as <h1> through <h6>, <p>, <ol>, <ul>, and <a>:

Property	Some Possible Values
text-align:	center, left, right, justify
text-decoration:	underline, line-through, blink
color:	blue, green, yellow, red, white, etc.
font-family:	Arial, Verdana, "Times New Roman"
font-size:	large, 120%, 20px (pixels)
font-weight:	bold, normal
font-style:	italic, normal

The actual list of available properties and values is quite long, but the ones listed above are the most common for formatting text via CSS.

How Browsers Process CSS

A web browser will process all CSS code it encounters, even if it is from all three methods.

For example, an external style sheet could define the font of a heading, an internal style sheet could specify the font size of the heading, and an inline style could italicize the heading. All three would be applied.

Sometimes a browser will receive conflicting instructions from the CSS code. For example, what if each of the above CSS sources specified a different color for the heading text?

Browsers need a consistent way of settling these formatting conflicts in a consistent fashion. That is where the "cascade" of cascading style sheets comes into effect.